

SIMATS ENGINEERING

AI-BASED SMART COLD CHAIN LOGISTICS MONITORING & OPTIMIZATION PLATFORM



**COLD CHAIN
INTELLIGENCE · IoT
+ AI PLATFORM · SRS
DOCUMENT**

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1. Problem Statement

Cold chain logistics is the temperature-controlled supply chain that keeps vaccines, pharmaceuticals, dairy products, seafood and fresh produce within a safe environmental range from the point of origin to the point of consumption. Any deviation in temperature, humidity or handling time along this chain directly degrades product quality, and in the case of medicines and vaccines, can render the product unsafe for use. Because these goods pass through multiple custodians — cold storage warehouses, refrigerated trucks, distribution hubs and last-mile delivery — the risk of an undetected excursion grows at every handover point.

Traditional cold chain monitoring still relies heavily on manual temperature logging, periodic spot-checks and isolated sensor readouts that are not connected to a central system. This gives stakeholders only a fragmented, delayed view of shipment conditions. Equipment such as refrigeration units and compressors are also inspected reactively — after a breakdown has already caused spoilage — rather than being monitored for early warning signs of failure. Delivery routes are planned with limited regard to real-time traffic, weather or refrigeration load, which increases transit time and the exposure window for temperature-sensitive cargo.

The consequence is a supply chain that is difficult to audit, slow to react to anomalies and expensive to operate: spoiled inventory, regulatory non-compliance, and lost customer trust. What the industry needs is a platform that continuously senses the physical condition of every shipment, reasons over that data using predictive models, and converts raw sensor streams into timely, actionable decisions — before spoilage occurs rather than after it is discovered.

2. Introduction

2.1 Purpose

This Software Requirement Specification (SRS) analyzes the cold chain logistics scenario described above and defines the functional scope, quality attributes, actors, use cases and constraints of the proposed AI-Based Smart Cold Chain Logistics Monitoring and Optimization Platform. It is written to guide the design, development and validation of the system, and to give reviewers a single reference for what the system must do and how well it must do it.

2.2 Business Domain

The system sits at the intersection of IoT sensing, predictive analytics and logistics operations, serving perishable-goods supply chains — pharmaceutical distribution, dairy and seafood logistics, and fresh-produce transport — where regulatory bodies (such as food-safety and drug-control authorities) mandate demonstrable, auditable temperature control.

2.3 Intended Audience

- Software architects and developers implementing the platform
- Fleet and warehouse operations teams who will use the system daily
- Quality assurance and compliance auditors validating cold chain integrity

3. Scope of the System

The platform covers the transportation and storage segment of the cold chain — from the moment goods are loaded into a monitored vehicle or cold room to the moment they are handed over at the destination. It ingests continuous data from IoT sensors, applies AI models to detect and predict anomalies, and surfaces the results through dashboards, alerts and reports. Warehouse-management, order-taking and billing functions of the wider business are outside this scope and are assumed to be handled by existing enterprise systems that the platform integrates with at the data level.

3.1 In Scope

- Continuous environmental sensing (temperature, humidity, GPS, door/shock events)
- Predictive maintenance for refrigeration units
- AI-assisted route optimization
- Automated, rule-driven and model-driven alerting
- Real-time operational dashboards and historical analytics
- Compliance-ready audit logging

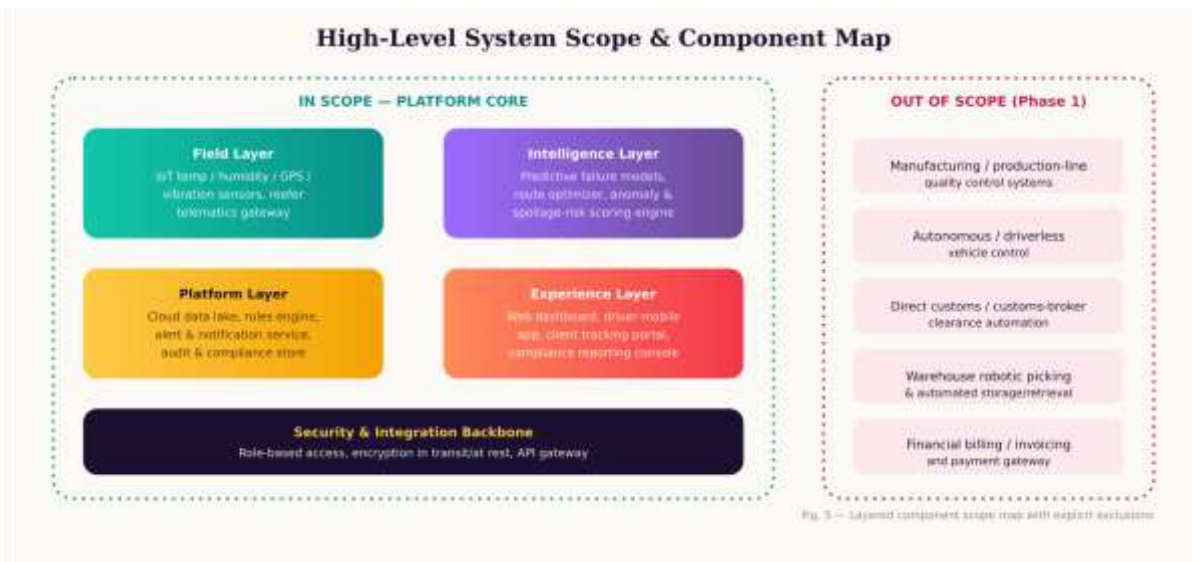
3.2 Out of Scope

- Order management, invoicing and payment processing
- Warehouse inventory control beyond cold-zone environmental status
- Vehicle manufacturing or hardware sensor design

4. Stakeholders and System Objectives

4.1 Stakeholders

Stakeholder	Interest in the System
Fleet Manager	Oversees vehicle assignment, monitors fleet-wide cold chain health, and acts on predictive-failure alerts.
Warehouse Operator	Monitors cold-storage zones and hands off shipments into the monitored transport network.
Delivery Driver	Follows optimized routes and receives in-transit alerts and delivery instructions.
Compliance Auditor	Reviews historical logs and compliance reports to verify regulatory adherence.
End Customer / Consignee	Receives goods and indirectly benefits from preserved product quality.
Platform Administrator	Configures sensor thresholds, manages users, and maintains system health.



Functional Requirements

Monitoring Module

FR-01

The system shall continuously capture temperature and humidity readings from IoT sensors at a configurable interval (default 60 seconds).

FR-02

The system shall capture and log real-time GPS location for every vehicle carrying an active shipment.

FR-03

The system shall detect and log door-open events and shock/vibration events that could indicate mishandling.

FR-04

The system shall allow authorized users to configure safe-range thresholds per product category (e.g., 2–8°C for vaccines).

4.2 Predictive Intelligence Module

FR-05

The system shall analyse refrigeration-unit sensor trends to predict impending equipment failure before a threshold breach occurs.

FR-06

The system shall estimate a Remaining Useful Life (RUL) indicator for each active reefer unit.

FR-07

The system shall flag anomalous sensor patterns (e.g., erratic compressor cycling) distinct from simple threshold breaches.

4.3 Route Optimization Module

FR-08

The system shall compute and recommend the optimal delivery route factoring live traffic, distance, and remaining cold-chain time budget.

FR-09

The system shall recalculate the route automatically when a significant deviation, delay, or road closure is detected.

FR-10

The system shall push updated route and ETA information directly to the driver's mobile application.

4.4 Alerting & Reporting Module

FR-11

The system shall generate a tiered automated alert (Info / Warning / Critical) whenever monitored conditions drift outside configured limits.

FR-12

The system shall route alerts to the correct stakeholder (driver, fleet manager, maintenance team) based on alert type and severity.

FR-13

The system shall provide a real-time dashboard summarizing the status of every shipment currently in transit.

FR-14

The system shall generate exportable compliance and quality reports for completed shipments.

4.5 User & Access Management

FR-15

The system shall support role-based login and access control for Fleet Managers, Drivers, Auditors, Warehouse Operators, and Clients.

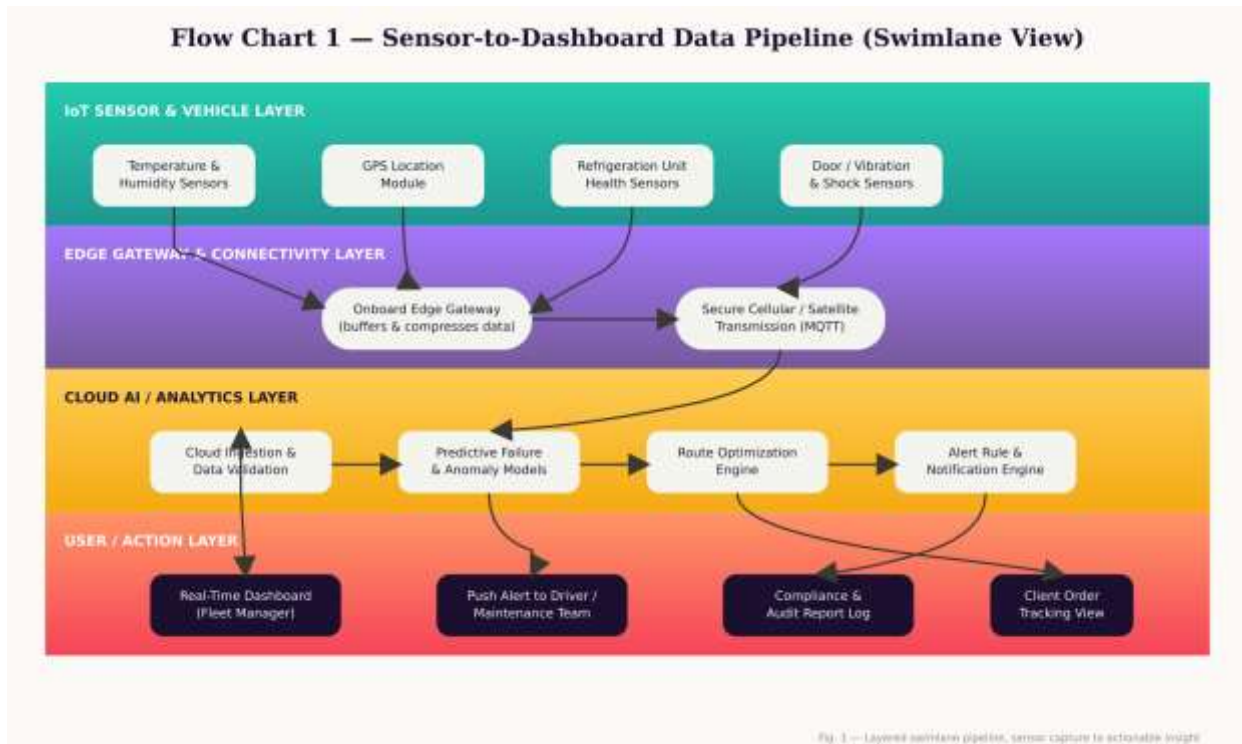
FR-16

The system shall allow Clients to track their own shipment and view its cold chain integrity record, without exposing other shipments.

Non-Functional Requirements

Quality Attribute	Requirement
Performance	Sensor-to-dashboard latency must not exceed 5 seconds under normal network conditions; the AI inference pipeline must score a new reading within 2 seconds.
Security	All telemetry must be encrypted in transit (TLS 1.2+) and at rest; access must be governed by role-based authentication.
Reliability	The platform must maintain 99.5% uptime for ingestion and alerting services, with automatic failover for the cloud ingestion layer.
Usability	Dashboards must be understandable by non-technical fleet staff within a 15-minute onboarding session.
Scalability	The architecture must scale horizontally to support at least a 10x increase in connected vehicles without redesign.

Quality Attribute	Requirement
Maintainability	AI models must be retrainable and redeployable without downtime to the ingestion or alerting services.
Availability	Edge gateways must buffer data locally during connectivity loss and resynchronize automatically once restored.
Interoperability	The platform must expose open APIs (REST/MQTT) so it can integrate with existing warehouse and ERP systems.



Actors and Use Cases

7.1 Primary Actors

- Fleet Manager — configures fleets, monitors overall cold chain health, receives escalated alerts.
- Warehouse Operator — monitors cold-storage zones and initiates shipment hand-off.
- Delivery Driver — follows optimized routes and receives in-transit guidance.
- Compliance Auditor — reviews logs and generates regulatory reports.

7.2 Secondary (System) Actor

- AI Prediction Engine — an automated actor that continuously scores telemetry for failure risk and feeds the route optimizer.

7.3 Use Case Diagram

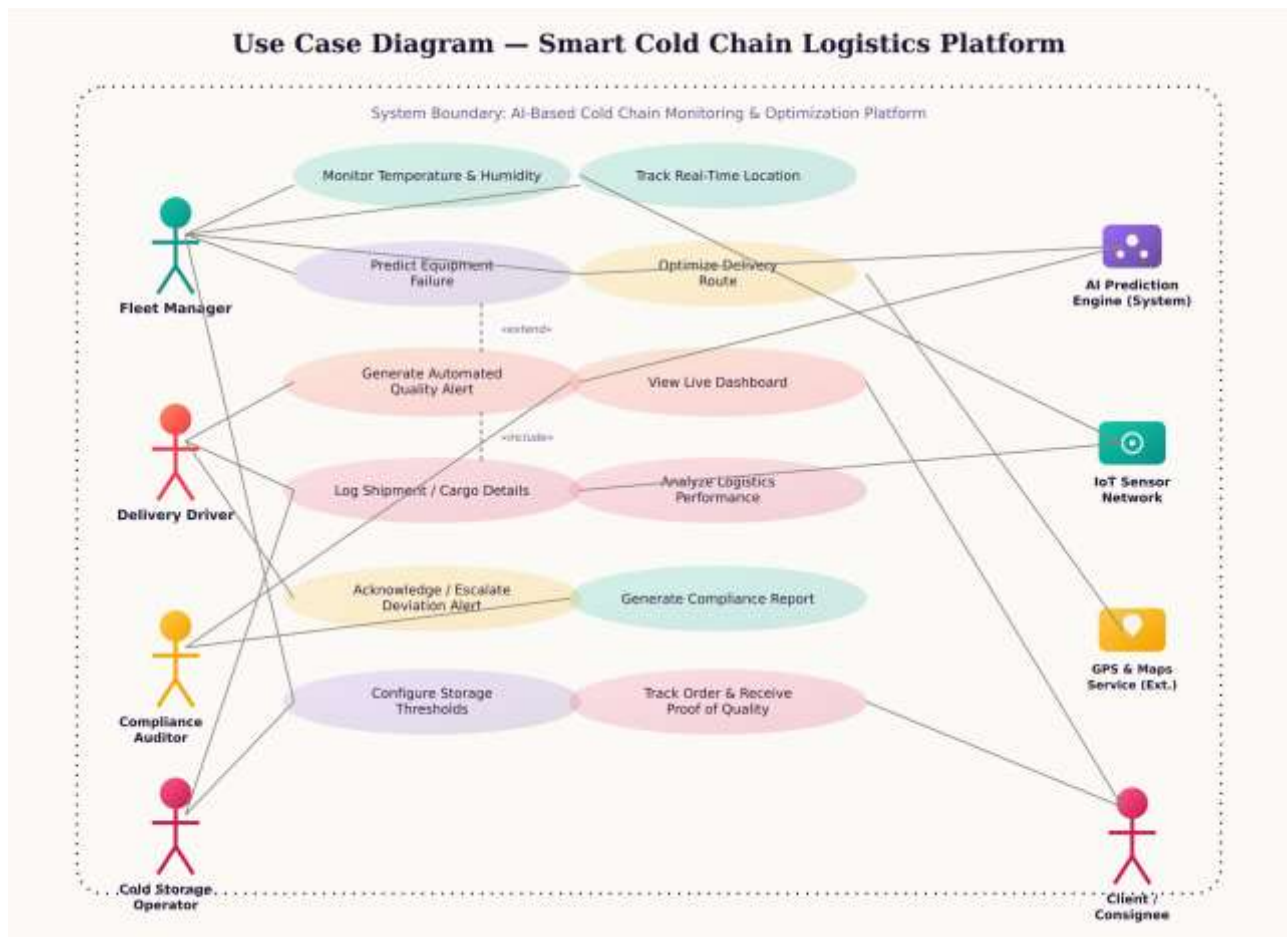


Figure 1 — Use case diagram: actors and their interactions with the Cold Chain Monitoring Platform.

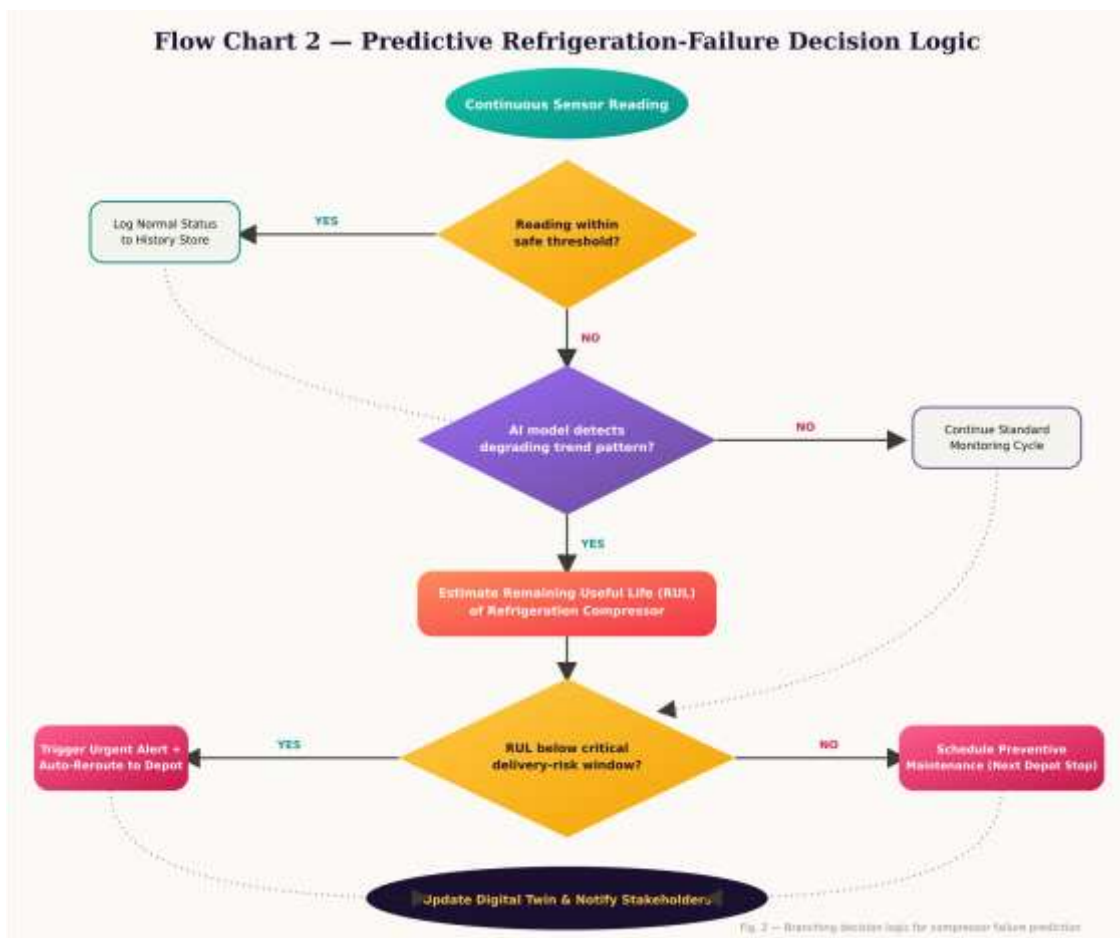
7.4 Use Case Descriptions

Use Case	Primary Actor	Description / Precondition
UC-01 Monitor Temperature & Humidity	Warehouse Operator	System streams live readings from active sensors. Precondition: sensors are paired and transmitting.
UC-02 Track Shipment Location	Fleet Manager	System plots the live GPS position of each vehicle on the dashboard map.
UC-03 Predict Equipment Failure	AI Prediction Engine	Model evaluates recent telemetry against learned failure patterns and raises a risk score.
UC-04 Optimize Delivery Route	AI Prediction Engine	System recomputes the route when traffic, weather or cold-exposure risk changes materially.
UC-05 Generate Automated Alert	Warehouse Operator / Driver	Triggered when a reading crosses a configured threshold; notifies the relevant role.

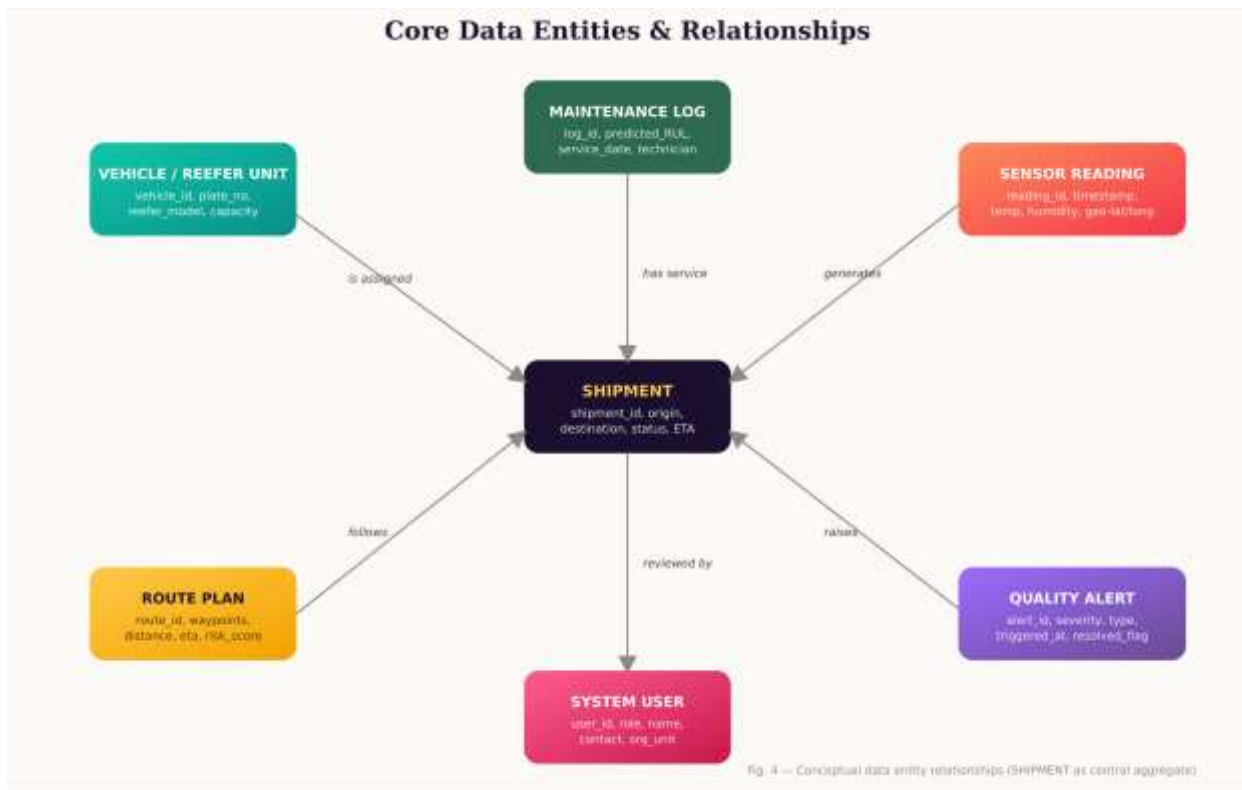
Use Case	Primary Actor	Description / Precondition
UC-06 View Real-Time Dashboard	Fleet Manager	Displays consolidated fleet status, active alerts and KPIs.
UC-07 Generate Compliance Report	Compliance Auditor	Compiles historical logs into a regulator-ready PDF/CSV report for a selected date range.

Flow chart:

The platform is organized into five cooperating layers: an IoT sensing layer at the edge, an edge gateway that buffers and filters data locally, a cloud ingestion layer, a central AI analytics engine, and a set of output channels that turn model results into human and machine action. A dashed feedback path lets the route optimizer push live corrections back to the driver's app without waiting for the next full analytics cycle.



9. Data Requirements



10. System Features — The Continuous Monitoring Cycle

Rather than behaving as a one-shot pipeline, the platform runs as a closed loop: every reading that enters the system eventually influences the next decision the system makes, and every decision changes what gets sensed next. The six stages below repeat continuously for as long as a shipment is in transit.



Figure — The continuous monitoring cycle underpinning every system feature.

#	Feature
1	Sensor Data Capture — raw environmental and location readings are generated at the edge.
2	Edge Filtering & Validation — noise is removed and readings are buffered if connectivity drops.
3	Cloud Feature Store — validated readings are time-stamped and stored for both real-time and historical use.
4	Predictive Model Inference — failure-risk and spoilage-risk models score the latest window of data.
5	Decision & Alert Engine — scores are compared against thresholds and business rules to decide on action.
6	Dashboard / Report Output — results reach the humans and systems that need to act on them.

11. Assumptions and Constraints

11.1 Assumptions

- All vehicles in scope can be fitted with a compatible IoT sensor and edge gateway.
- A cellular or LoRaWAN network is available along the majority of transport routes.
- Historical sensor and failure data is available in sufficient volume to train the prediction models.
- Stakeholders will complete role-based onboarding before receiving system access.

11.2 Constraints

Type	Constraint
Technical	Edge devices have limited compute and must run lightweight inference or none at all, deferring heavy models to the cloud.
Operational	Field sensors must survive vehicle vibration, humidity and a wide temperature range (-25°C to 50°C).
Economic	Sensor and connectivity cost per vehicle must stay within the project's per-unit budget cap.
Legal / Regulatory	Audit logs must satisfy retention and traceability rules mandated for pharmaceutical and food transport.
Time	The platform must be pilot-ready within one operational quarter to align with the sponsoring fleet's rollout plan.

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12.1 Suggested Improvements

- Add a formal data-retention policy requirement once the specific regulatory jurisdiction is confirmed.
- Introduce a simulation/sandbox mode so new thresholds can be validated before being applied to live shipments.

- Extend the use case model with an exception flow for total connectivity loss beyond the edge buffer's capacity.

- **Shipment**

Central record: identifiers, origin/destination, product category, current status, ETA. Retained for the regulatory-mandated audit period (typically 3–5 years for pharma cargo).

- **Sensor Reading**

High-volume time-series data: timestamp, temperature, humidity, GPS coordinates. Stored in a time-series optimized store; raw data aggregated after 90 days.

- **Vehicle / Reefer Unit**

Master data: vehicle ID, reefer model, capacity, current health indicators.

- **Quality Alert**

Severity, type, trigger time, resolution status — immutable once logged, to preserve audit integrity.

- **Route Plan**

Waypoints, distance, ETA, computed spoilage-risk score for the active leg.

- **Maintenance Log**

Predicted RUL, service dates, technician notes — links predictive output to real-world action.

- **System User**

Role-based identity record used for access control and alert routing.

Conclusion

The requirement analysis presented in this document translates the cold chain logistics problem statement into a concrete, testable set of functional and non-functional requirements, grounded in clearly identified actors, use cases and system architecture. By closing the loop between sensing, prediction and action, the proposed platform is positioned to materially reduce spoilage, strengthen compliance, and give every stakeholder — from driver to auditor — a shared, real-time view of cold chain health.